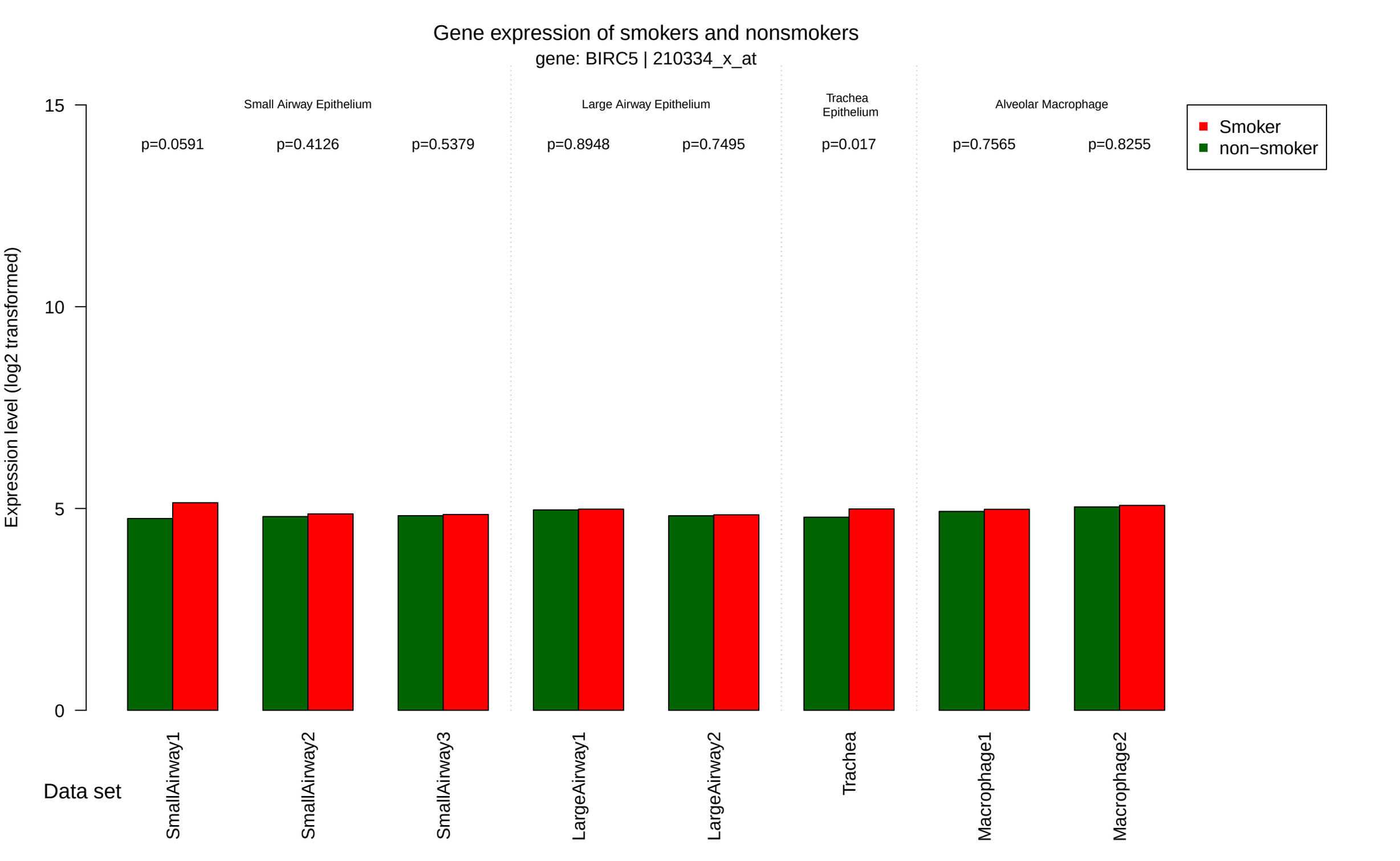
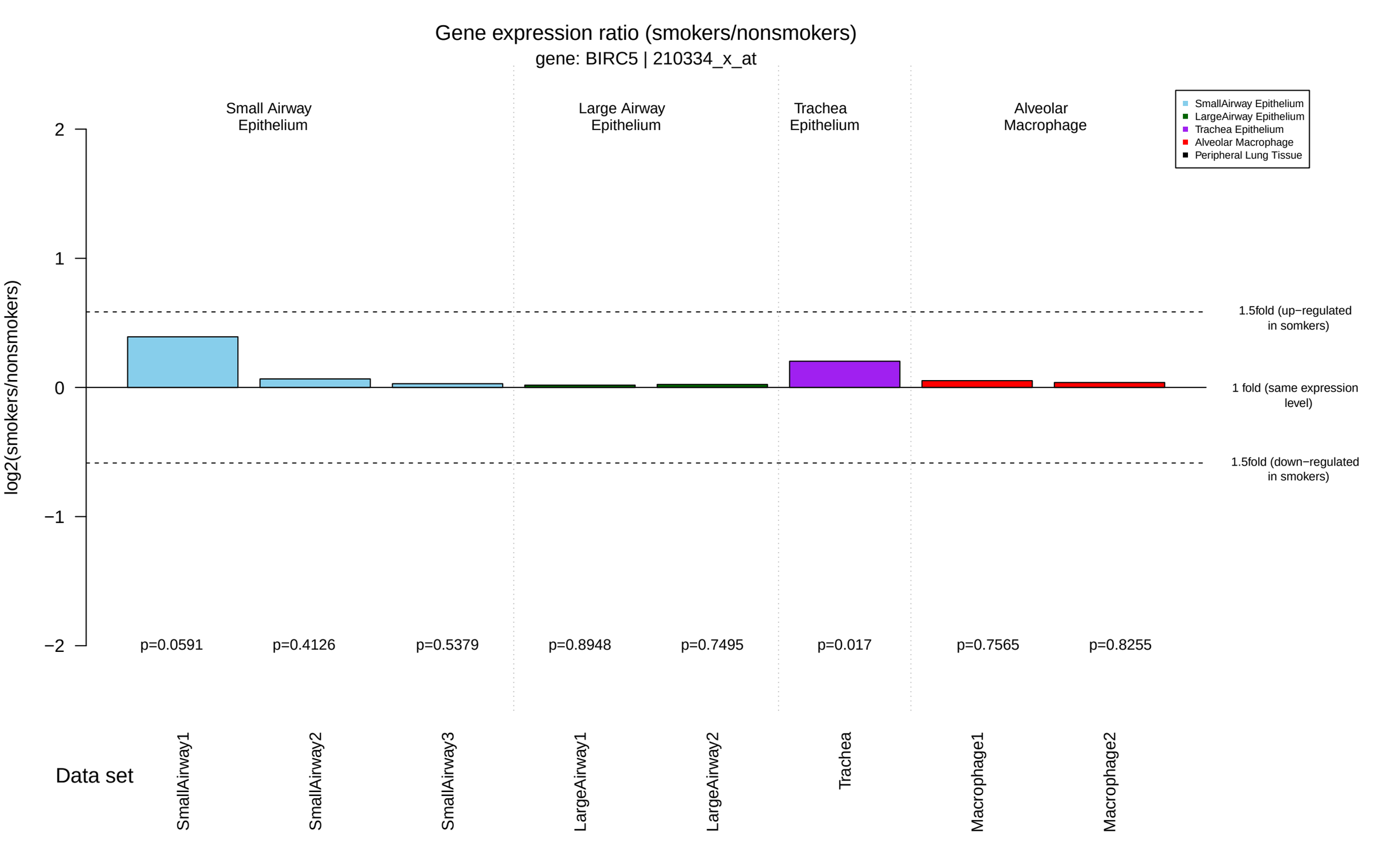
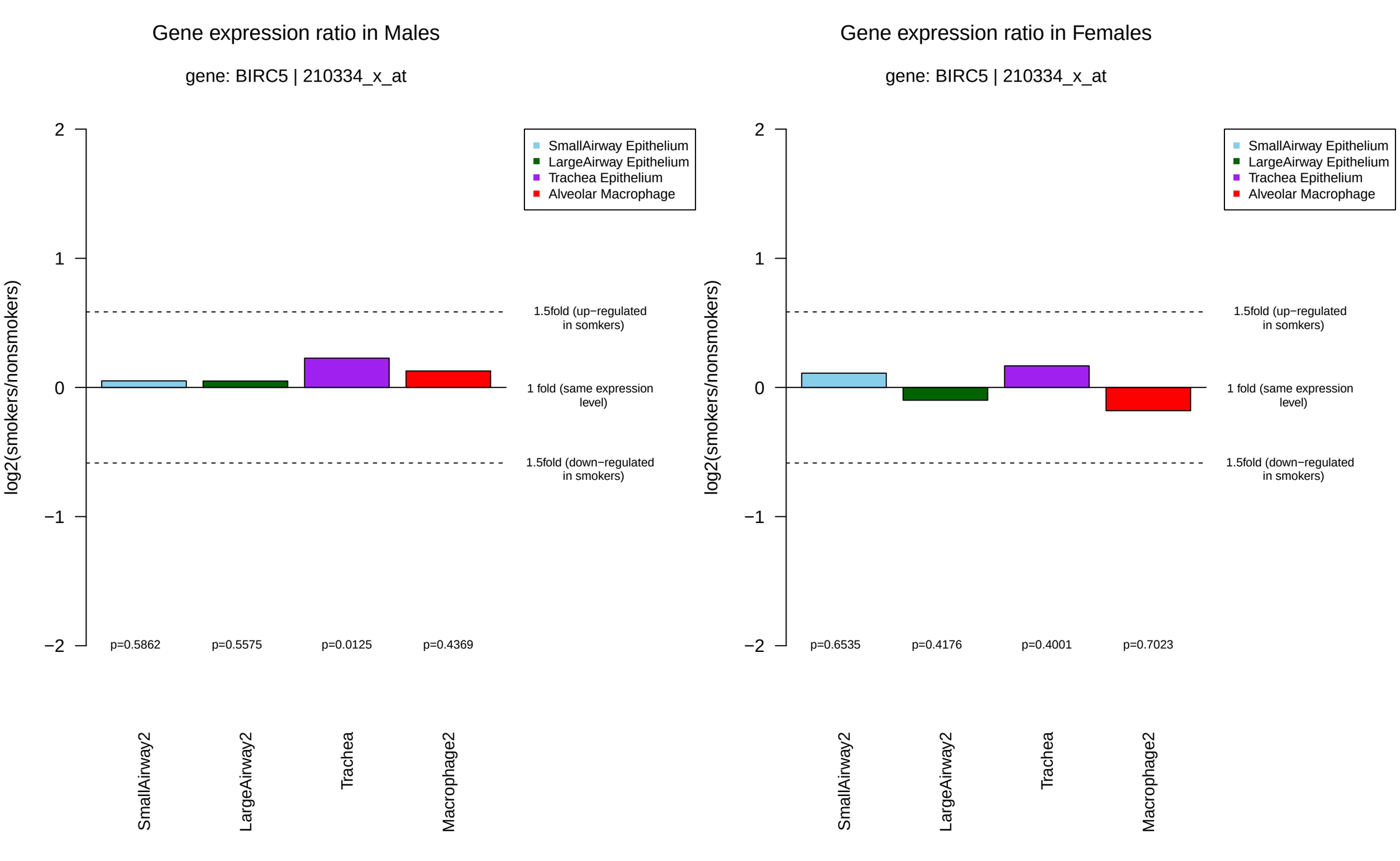
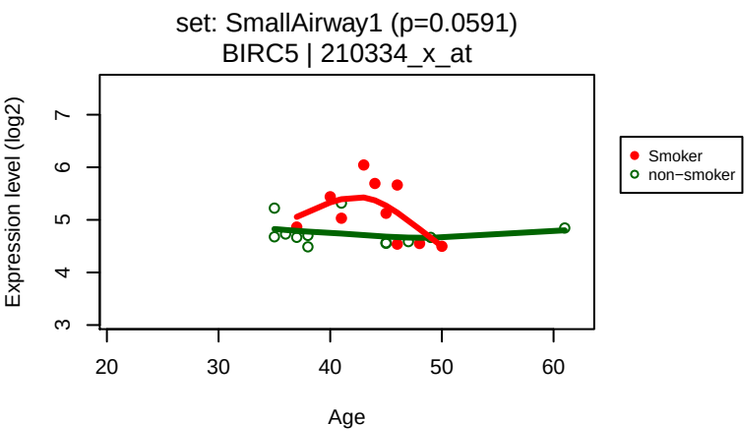
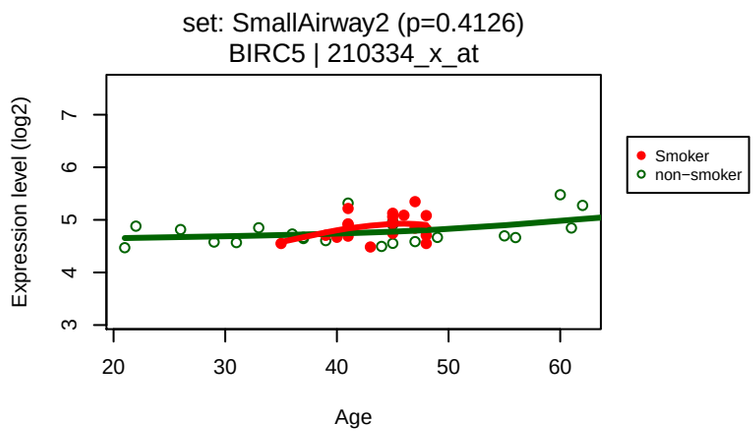
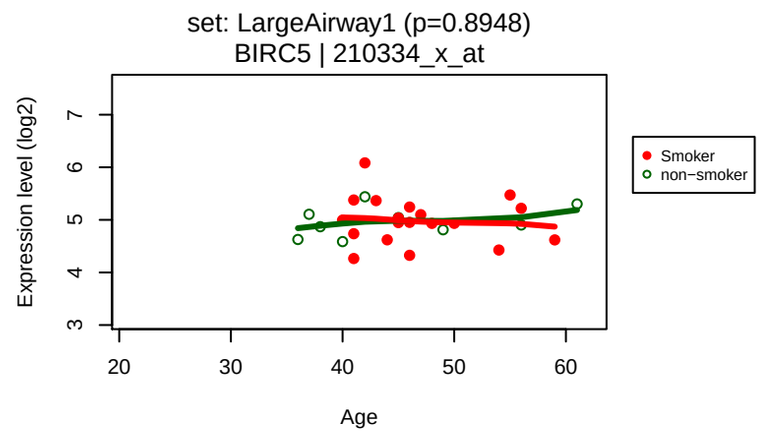

Gene expression of smokers and nonsmokers
gene: BIRC5 | 210334_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0591
p=0.4126
p=0.5379
p=0.8948
p=0.7495
p=0.017
p=0.7565
p=0.8255
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

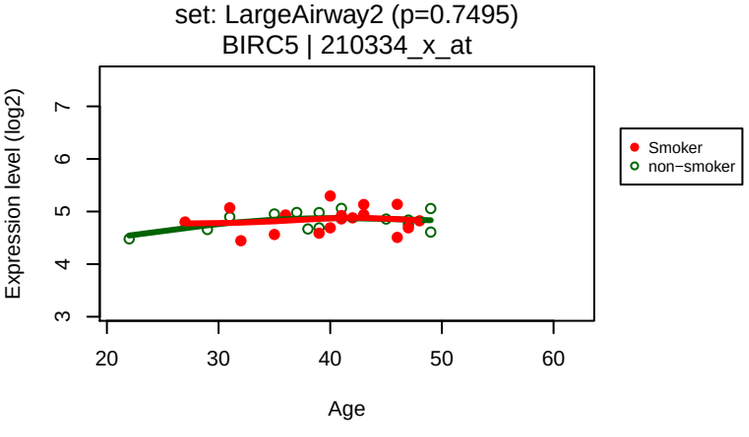

Gene expression ratio (smokers/nonsmokers)
gene: BIRC5 | 210334_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0591
p=0.4126
p=0.5379
p=0.8948
p=0.7495
p=0.017
p=0.7565
p=0.8255
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

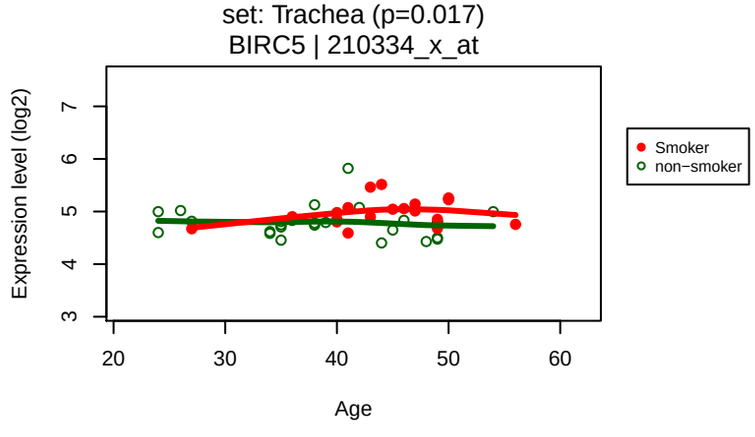

Gene expression ratio in Males
gene: BIRC5 | 210334_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5862
p=0.5575
p=0.0125
p=0.4369
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BIRC5 | 210334_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6535
p=0.4176
p=0.4001
p=0.7023
SmallAirway2
LargeAirway2
Trachea
Macrophage2

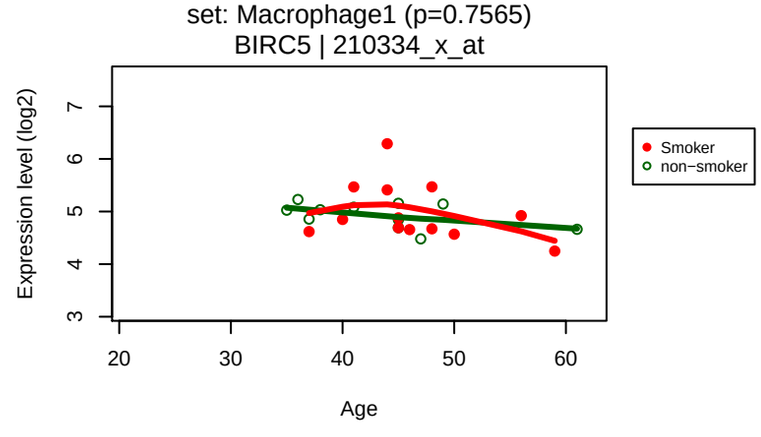

set: SmallAirway1 (p=0.0591)
BIRC5 | 210334_x_at
Expression level (log2)
Age
Smoker
non−smoker

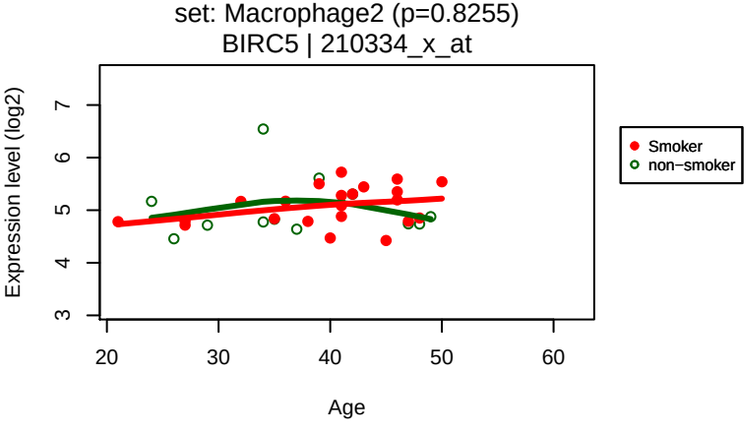

set: SmallAirway2 (p=0.4126)
BIRC5 | 210334_x_at
Expression level (log2)
Age
Smoker
non−smoker

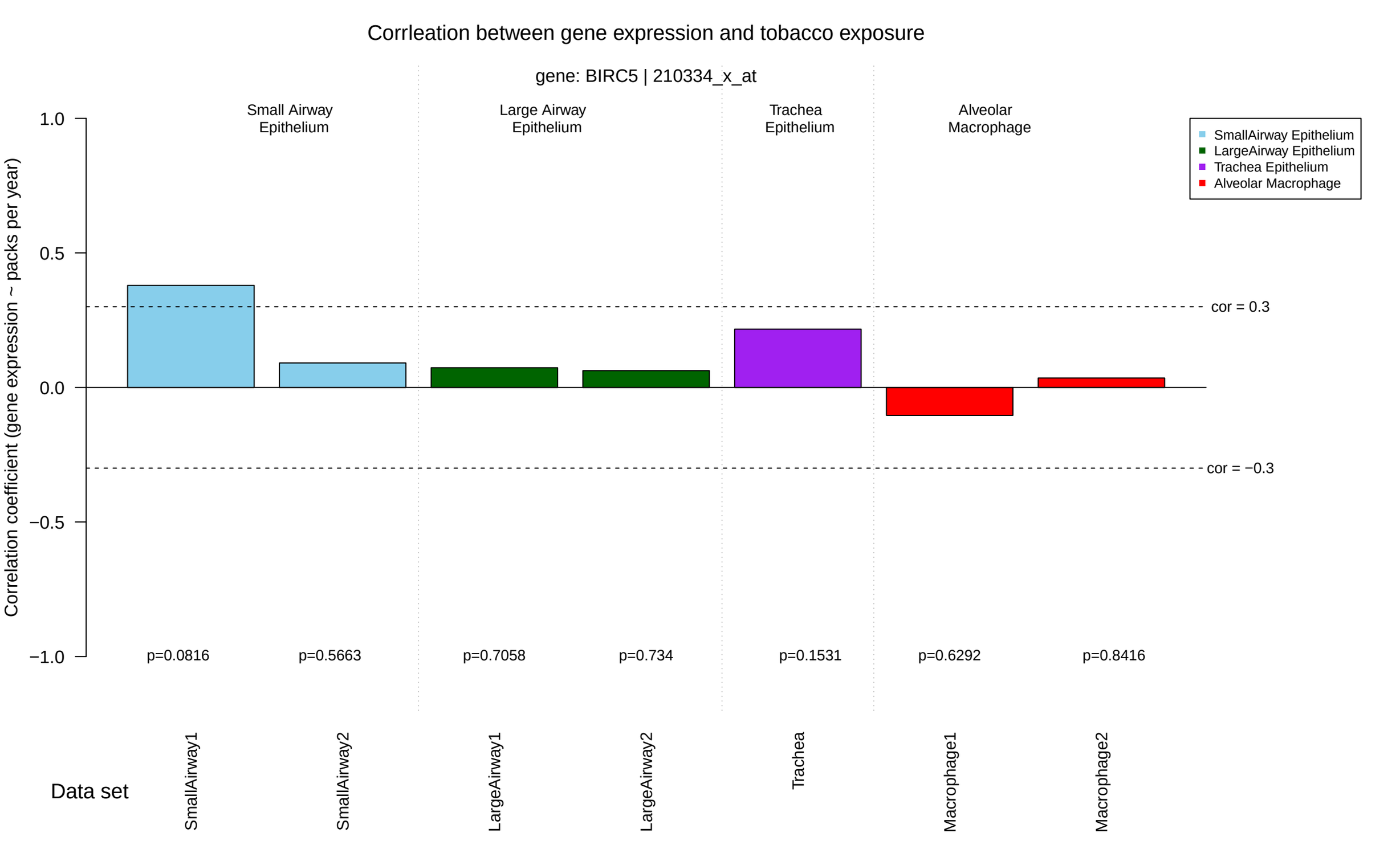

set: LargeAirway1 (p=0.8948)
BIRC5 | 210334_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7495)
BIRC5 | 210334_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.017)
BIRC5 | 210334_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7565)
BIRC5 | 210334_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8255)
BIRC5 | 210334_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BIRC5 | 210334_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0816
p=0.5663
p=0.7058
p=0.734
p=0.1531
p=0.6292
p=0.8416
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2